Expt. No. 3

Title: Design and verify the layout of NMOS and PMOS devices in micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

Experiment No. 3

Aim: Design and verify the layout of NMOS and PMOS devices in micro-wind.

Software Required: Micro-wind software

Hardware: computer

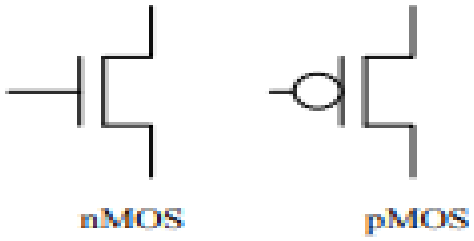
Theory:

NMOS

NMOS uses N-type FET to implement logic gates and other digital circuits. This NMOS transistor operates by creating an inversion layer in an P-type transistor body. The inversion layer called the n channel can conduct electrons between n and p-type source and drain.

PMOS

P-type MOS logic uses p channel metal oxide semiconductor FET to implement logic gates and other digital circuits. A PMOS transistor operates by creating an inversion layer in an N-type transistor body. This inversion layer called the P-channel, can conduct holes between P-type source and drain terminals. The strong logic is logic 1.



Lay Out Design Rules:

The design rules are usually described in two ways:

Micron rules, in which the layout constraints such as minimum feature sizes and minimum allowable feature separations, are stated in terms of absolute dimensions in micrometres or Lambda rules, which specify the layout constraints in terms of a single parameter (λ) and thus, allow linear, proportional scaling of all geometrical constraints.

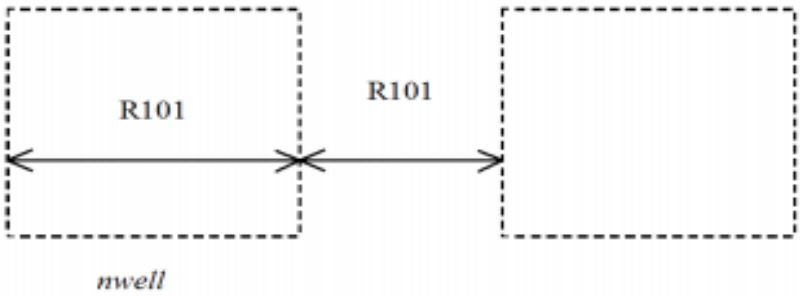
Lambda-based layout design rules were originally devised to simplify the industry-standard micron-based design rules and to allow scaling capability for various processes. It must be emphasized, however, that most of the submicron CMOS process design rules do not lend themselves to straightforward linear scaling. The use of lambda-based design rules must therefore be handled with caution in submicron geometries

N-Well

r101 Minimum well size 12 λ

r102 Between wells 12 λ

r110 Minimum well area 144 λ 2



Diffusion

r201 Minimum N+ and P+ diffusion width 4 λ

r202 Between two P+ and N+ diffusions 4 λ

r203 Extra n-well after P+ diffusion : 6 λ

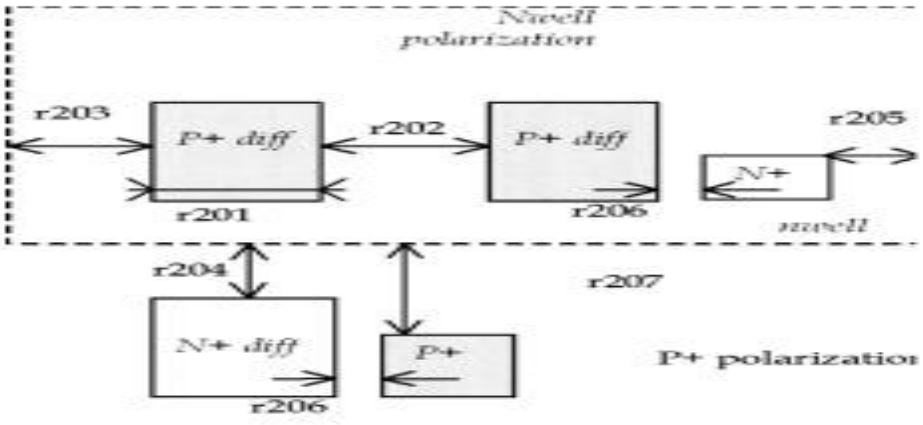
r204: Between N+ diffusion and n-well 6 λ

r205 Border of well after N+ polarization 2 λ

r206 Between N+ and P+ polarization 0 λ

r207 Border of N-well for P+ polarization 6 λ

r210 Minimum diffusion area 24 λ 2



Polysilicon

r301 Polysilicon Width 2 λ

r302 Polysilicon gate on diffusion 2 λ

r303 Polysilicon gate on diffusion for high voltage MOS 4 λ

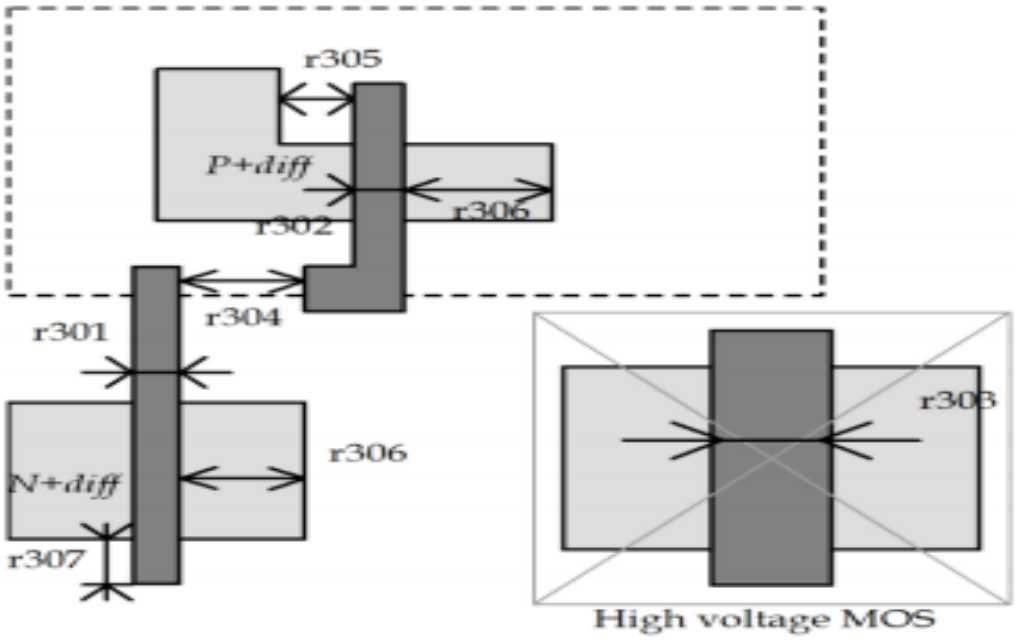
r304 Between two polysilicon boxes 3 λ

r305 Polysilicon Vs. other diffusion 2 λ

r306 Diffusion after polysilicon 4 λ

r307 Extra gate after polysilicon 3 λ

r310 Minimum surface 8 λ 2

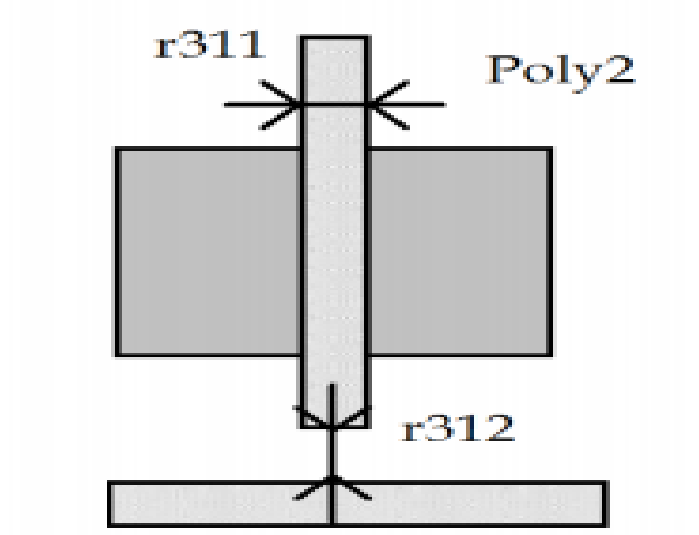


2nd Polysilicon Design Rules

r311 Polysilicon2 with 2 λ

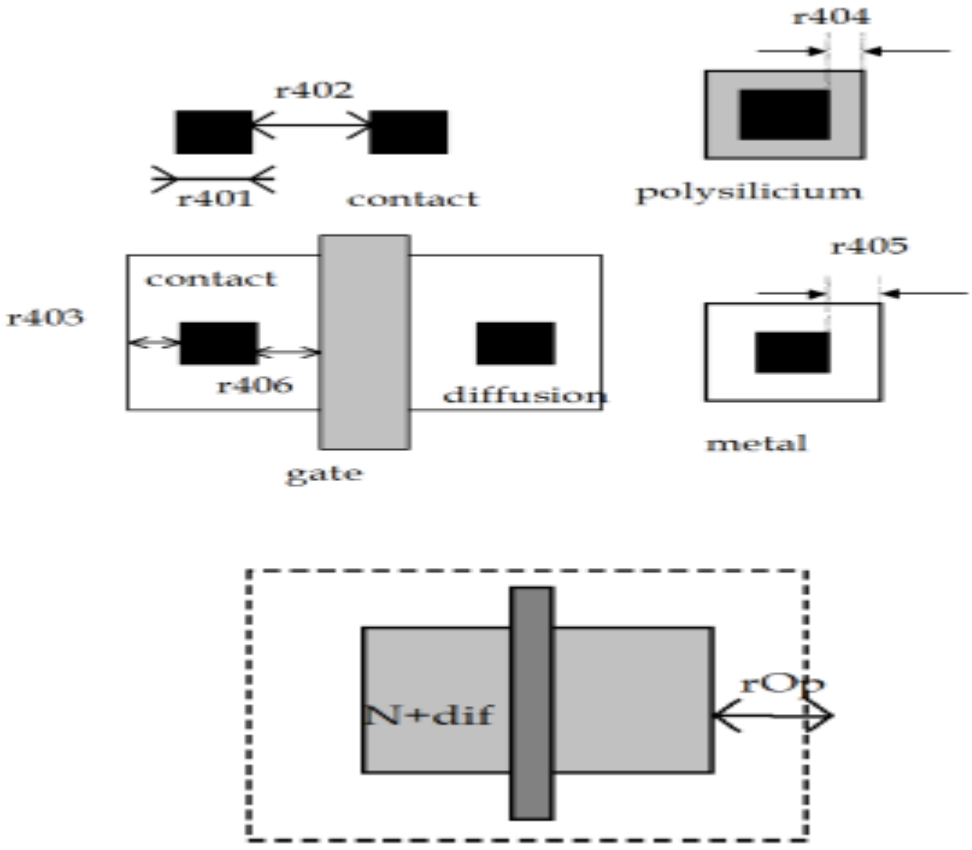
r312 Polysilicon gate on diffusion 2 λ

r320 Polysilicon2 minimum surface 8 λ



MOS Option

rOp Border of "option" layer over diff 7 N+ and diff P+



Contact

r401 Contact width 2λ

r402 Between two contacts 5λ

r403 Extra diffusion over contact 2λ

r404 Extra poly over contact 2λ

r405 Extra metal over contact 2λ

r406 Distance between contact and poly gate 3λ

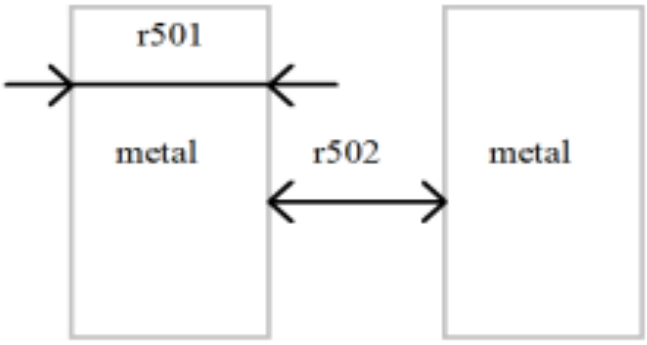
r407 Extra poly2 over contact 2λ

Metal 1

r501 Metal width 2λ

r502 Between two metals 2λ

r510 Minimum surface 16λ2



Via

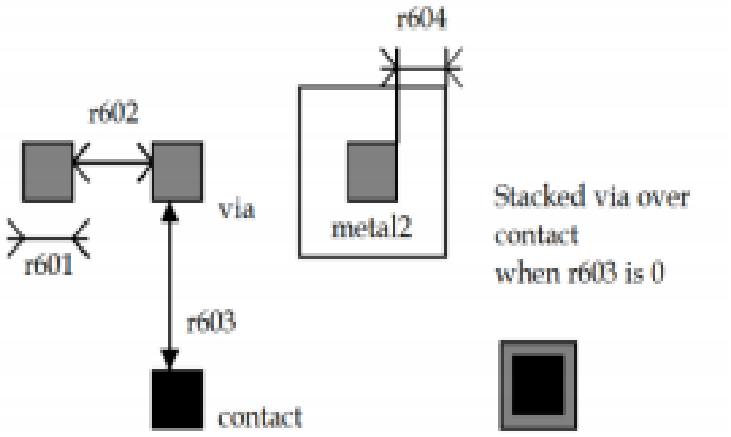
r601 Via width 2λ

r602 Between two Via 5λ

r603 Between Via and contact 0λ

r604 Extra metal over via 2λ

r605 Extra metal2 over via: 2λ

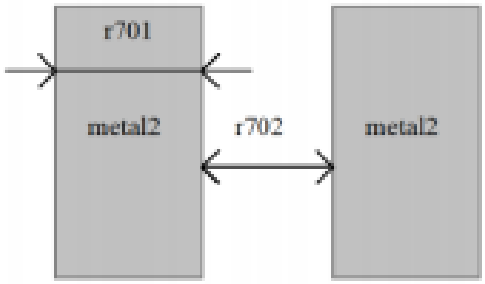


Metal 2

r701 Metal width: 4λ

r702 Between two metal2 4λ

r710 Minimum surface 16λ2



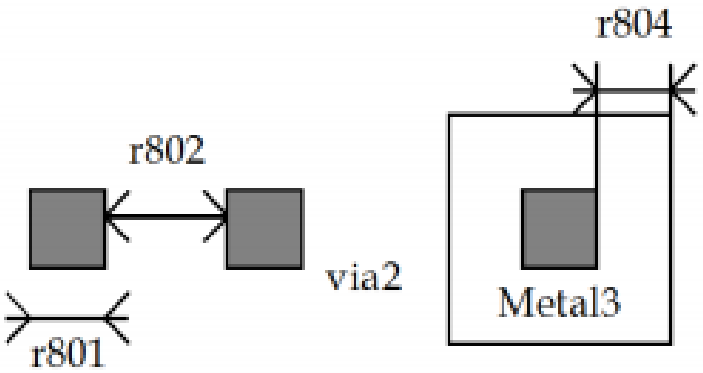
Via 2

r801 Via 2 width: 2λ

r802 Between two Via2: 5λ

r804 Extra metal2 over via2: 2λ

r805 Extra mental3 over via2: 2λ

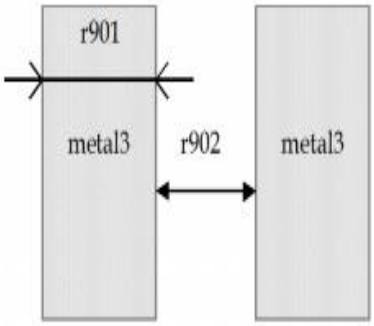


Metal 3

r901 Metal3 width: 4λ

r902 Between two metal3: 4λ

r910 Minimum surface: 32λ 2



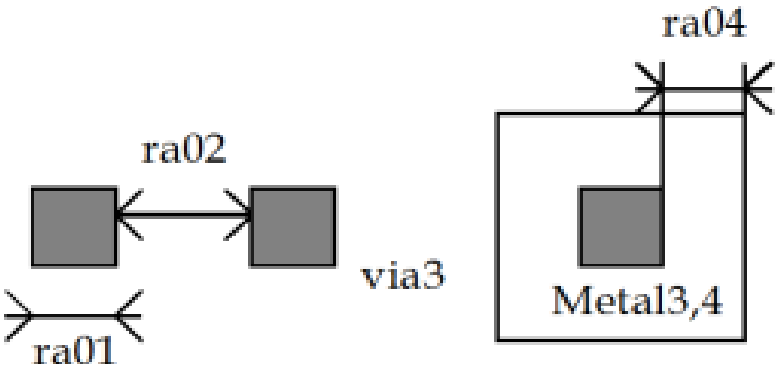
Via 3

ra01 Via3 width : 2λ

ra02 Between two Via3 : 5λ

ra04 Extra metal3 over via3 : 2λ

ra05 Extra metal4 over via3 : 2λ



Pads

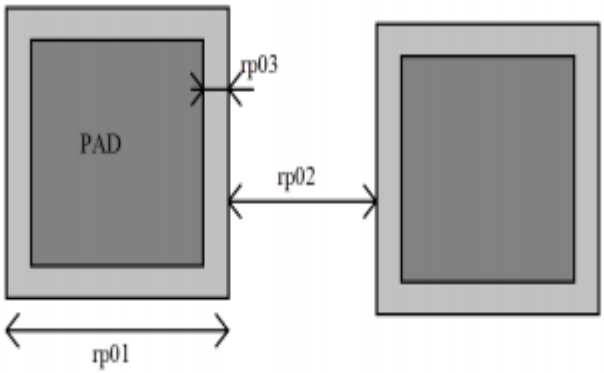
rp01 Pad width : 10µm

rp02 Between two pads 100µm

rp03 opening in passivation v.s via 5µm

rp04 opening in passivation v.s metals : 5µm

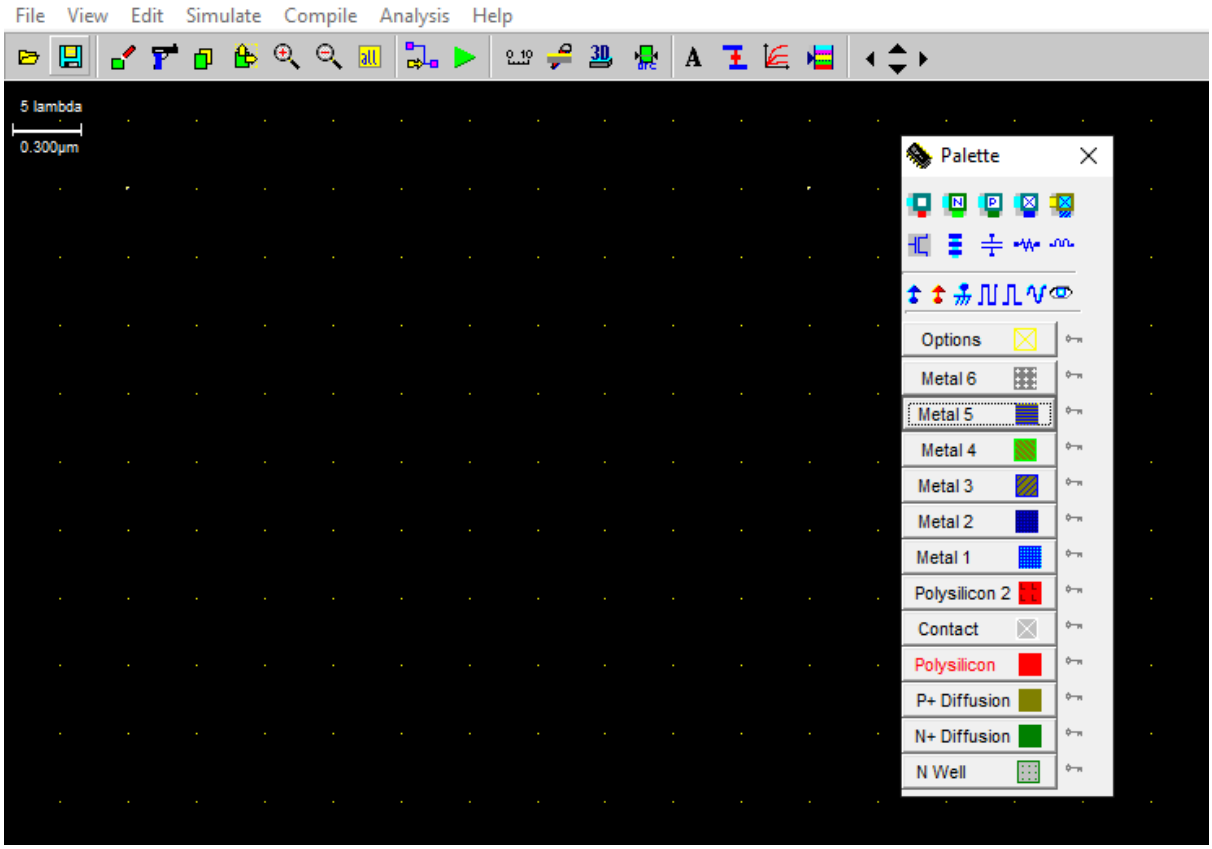
rp05 Between pad and unrelated active area : 20µm

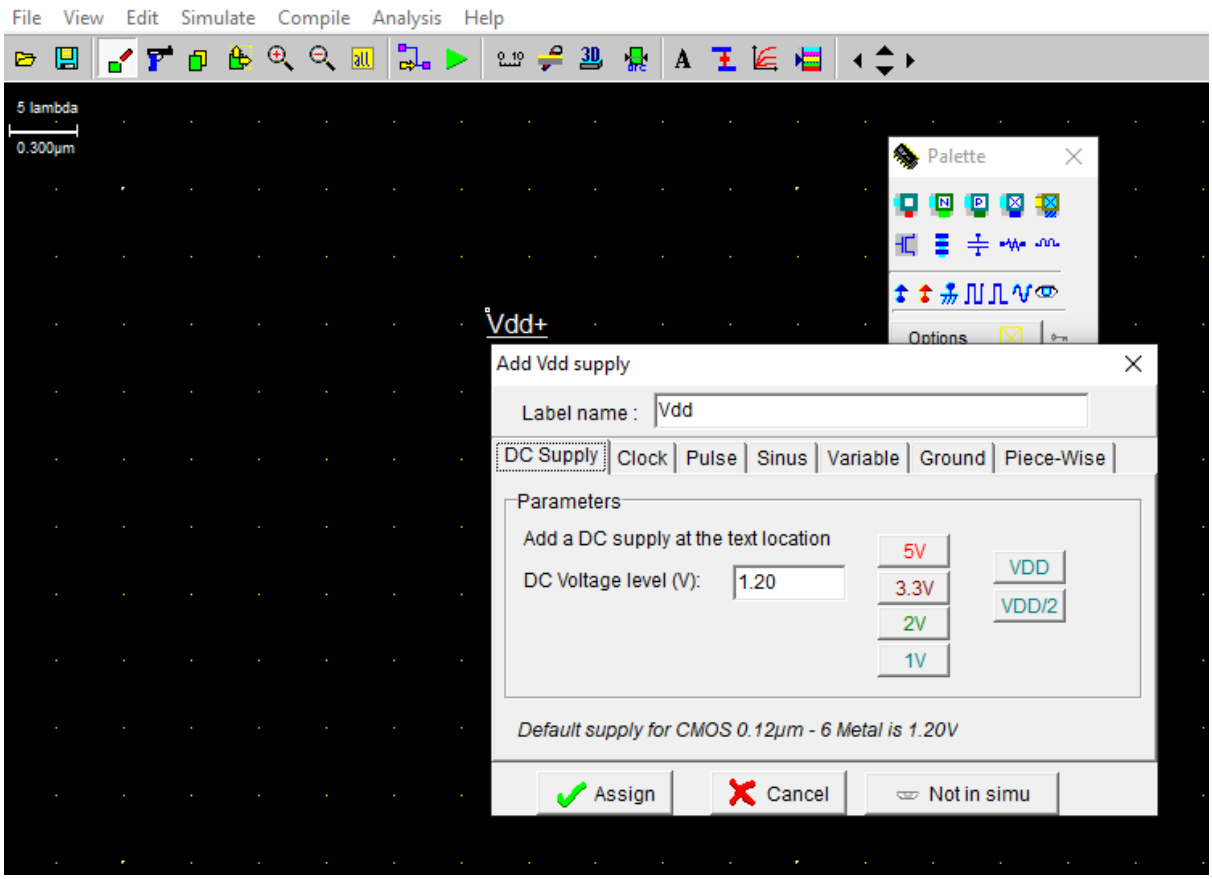


MOS layout

We use Microwind3.1 to draw the MOS layout and simulate its behavior. Go to the directory in which the software has been copied (By default microwind3). Double-click on the Microwind3.1 icon. The Microwind3.1 display window includes four main windows: the main menu, the lay out display window, the icon menu and the layer palette. The lay out window features a grid, scaled in lambda units. The lambda unit is fixed to half of the

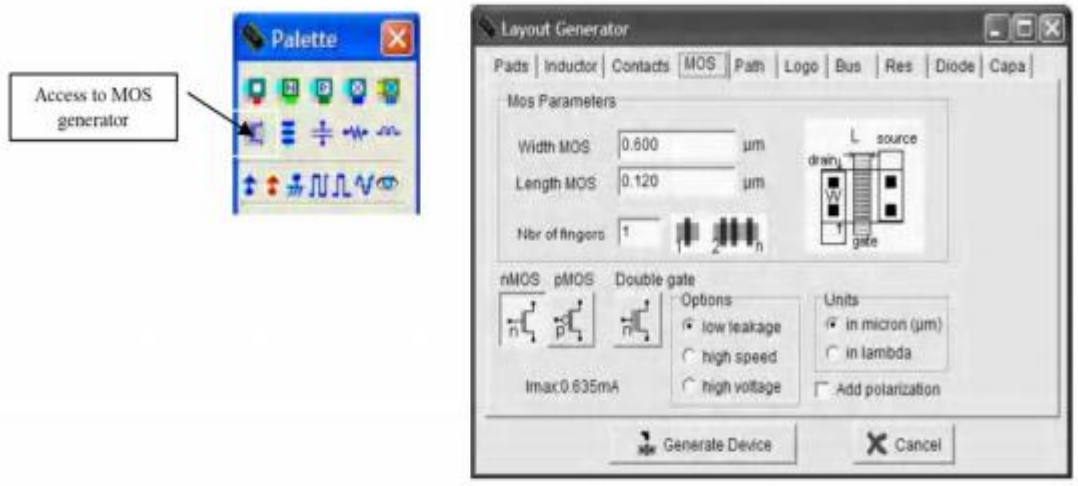
minimum available lithography of the technology. The default technology is a CMOS 6-metal layers 0.12µm technology, consequently lambda is 0.06µm (60nm).





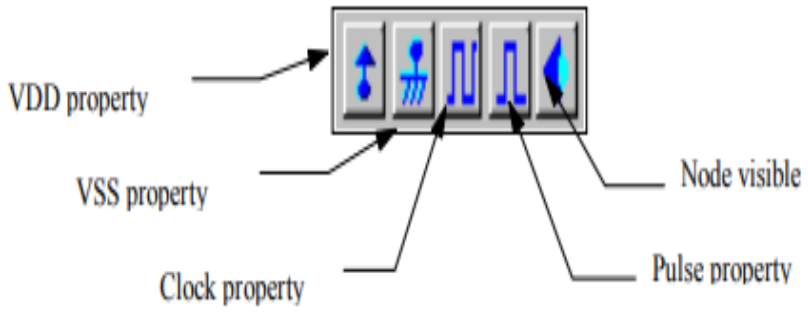
Layout considerations

The safest way to create a MOS device is to use the MOS generator. In the palette, click the MOS generator icon. A window appears as reported below. The programmable parameters are the MOS width, length, the number of gates in parallel and the type of device (n-channel or p-channel). By default metal interconnects and contacts are added to the drain and source of the MOS. We may add a supplementary metal2 interconnect on the top of metal 1 for drain and source

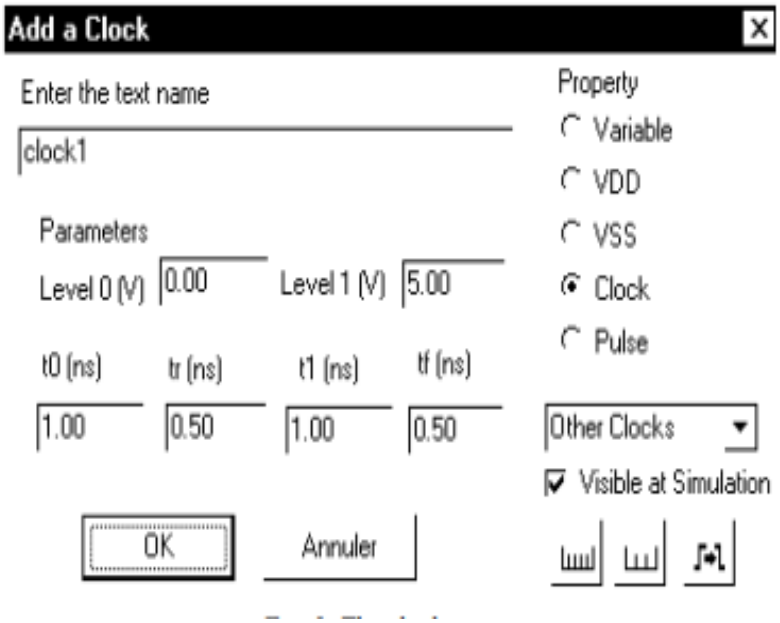


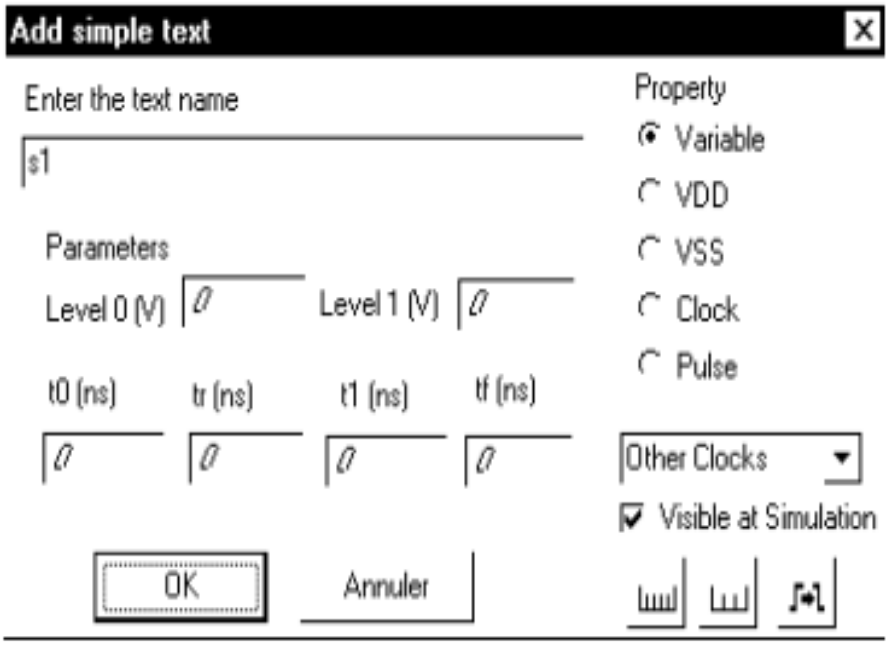
Add Properties for Simulation

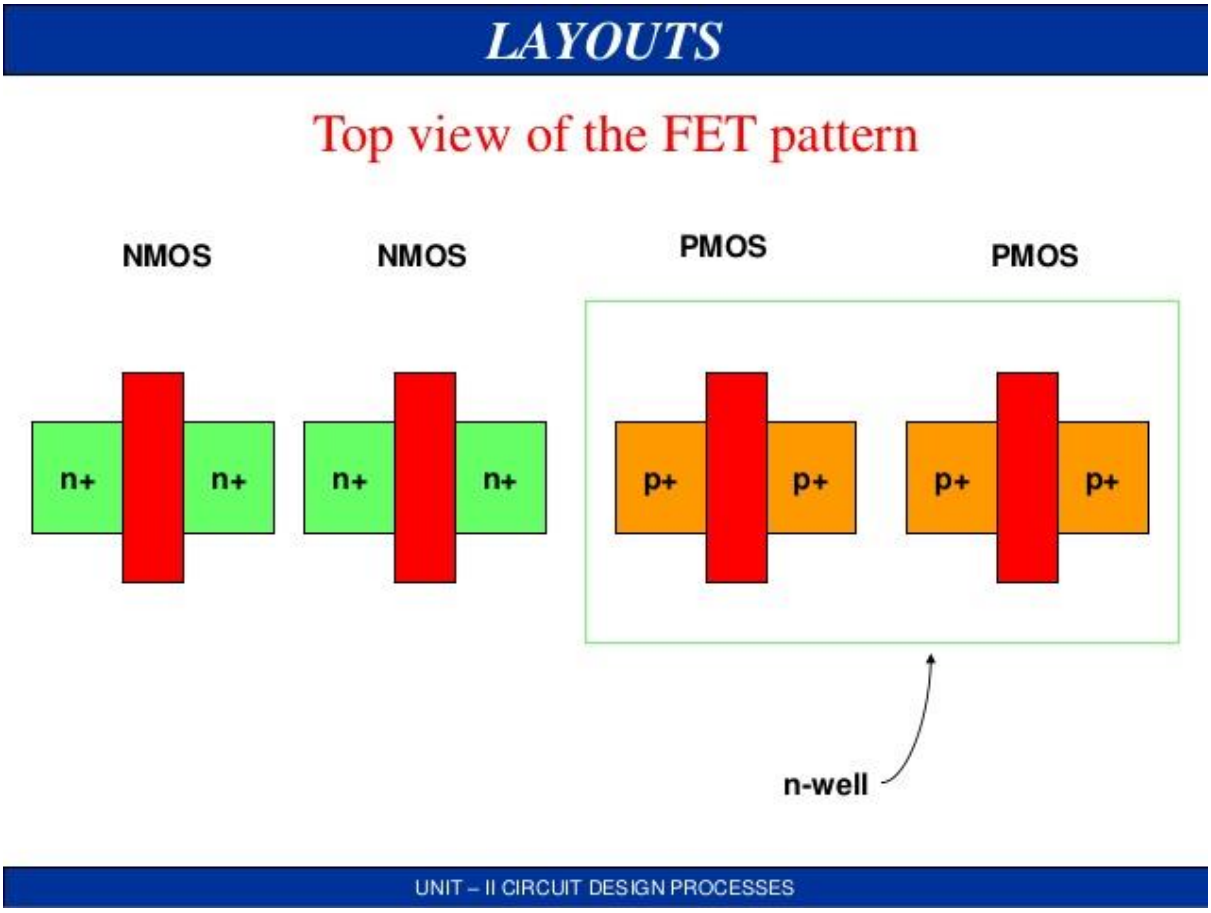
Properties must be added to the layout to fix the ground, the supply, the input and the outputs. The summary of available properties is reported below.



Stuck at Vdd: Click on the Vdd icon and click on the upper p-diffusion box. The Vdd property is sent to the node. Page 6 o Also click in the N-Well region inside which the pMOS is located. The NWell must always be at VDD voltage to keep the PMOS junctions inverted. p Hold at Vss: Click on the Vss icon, click on the lower n-diffusion box. The Vss property is sent to the node. q Apply a clock to node IN: Click on the Clock icon and on the polysilicon gate. The Clock menu appears (See below). Click on OK. A default clock with 3 ns period is generated. The Clock property is sent to the node and appears at the right hand side of the desired location with the name « clock1 ».


Add a Clock
Enter the text name
clock1
Parameters
Level 0 (V)
0.00
Level 1 (V)
5.00
t0 (ns)
tr (ns)
t1 (ns)
tf (ns)
1.00
0.50
1.00
0.50
Property
Variable
VDD
VSS
Clock
Pulse
Other Clocks
Visible at Simulation
OK
Annuler


Add simple text
Enter the text name
$1
Parameters
Level 0 (V)
0
Level 1 (V)
0
t0 (ns)
tr (ns)
t1 (ns)
tf (ns)
0
0
0
0
Property
Variable
VDD
VSS
Clock
Pulse
Other Clocks
Visible at Simulation
OK
Annuler



Procedure:

1) Start micro-wind software.

2) Design the PMOS and NMOS devices following the layout design.

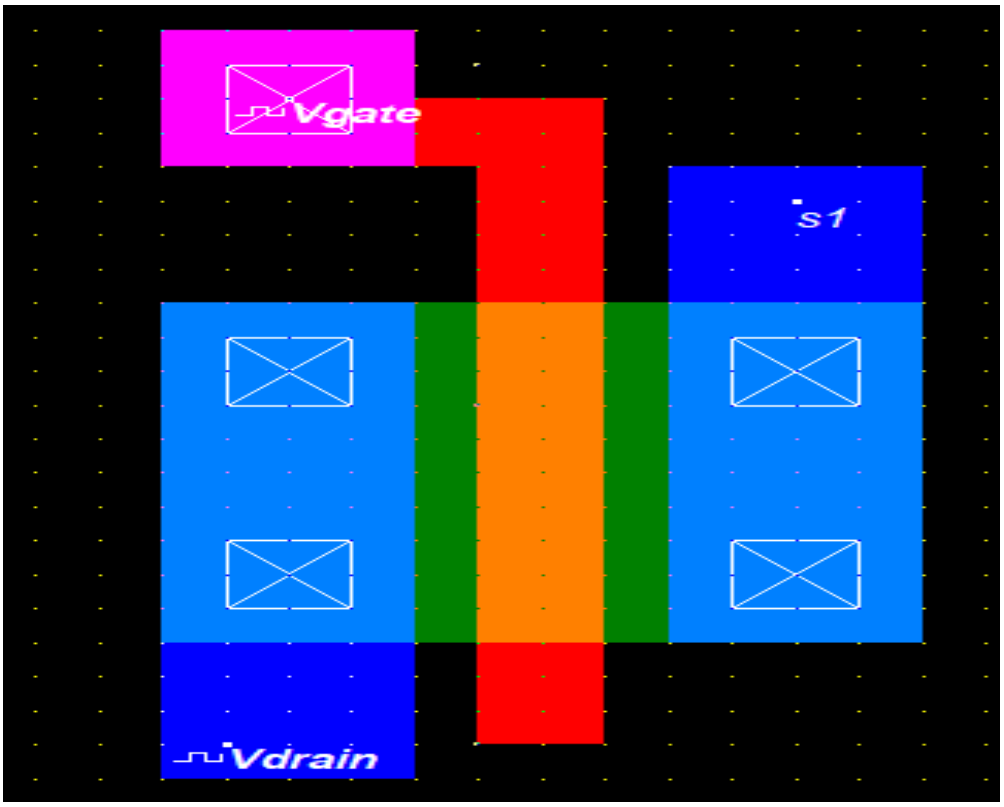
3) Click on DRC at every stage of design.

4) Save your design and finally run DRC again.

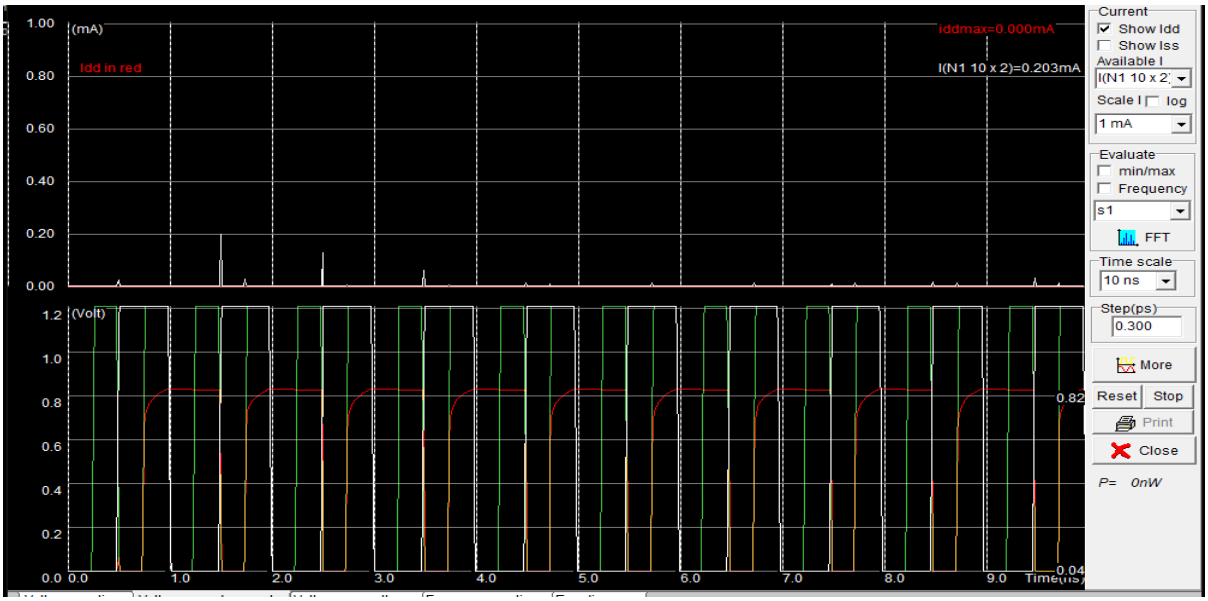
5) Observe various outputs using the stimulation option

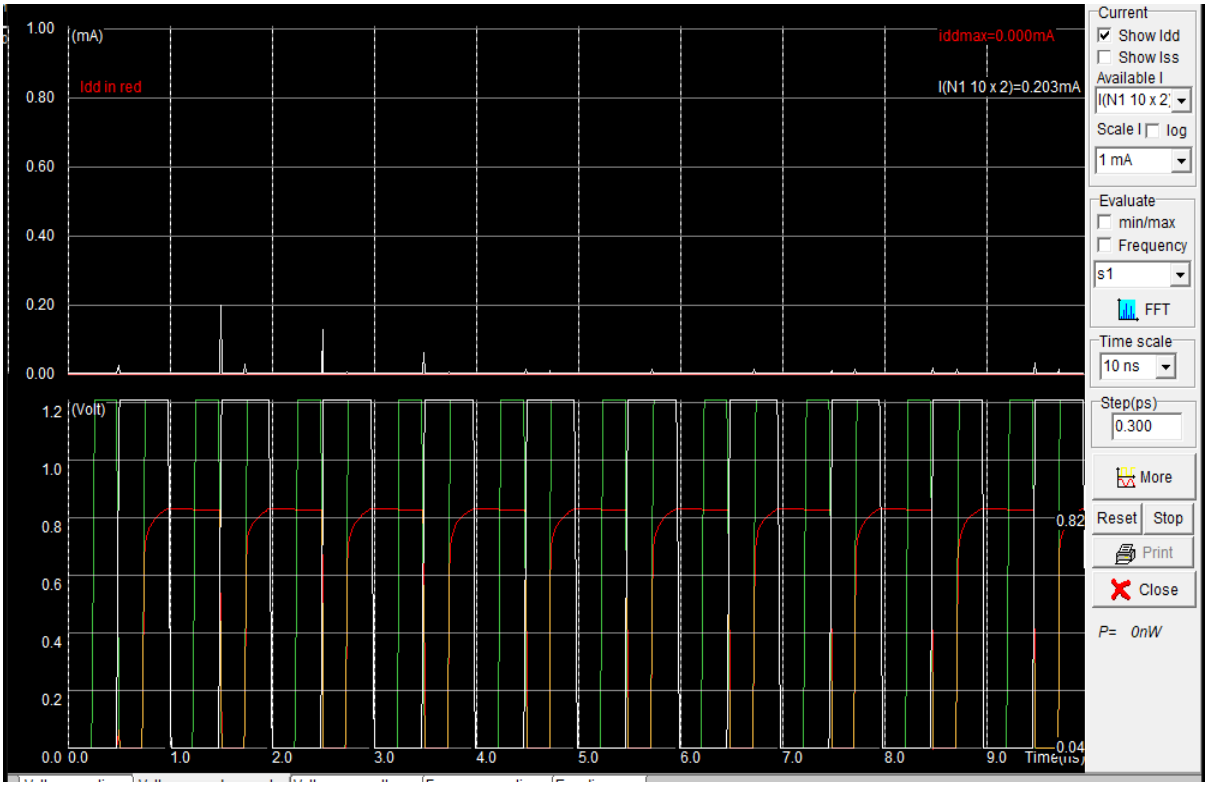
Layout:

N-MOS

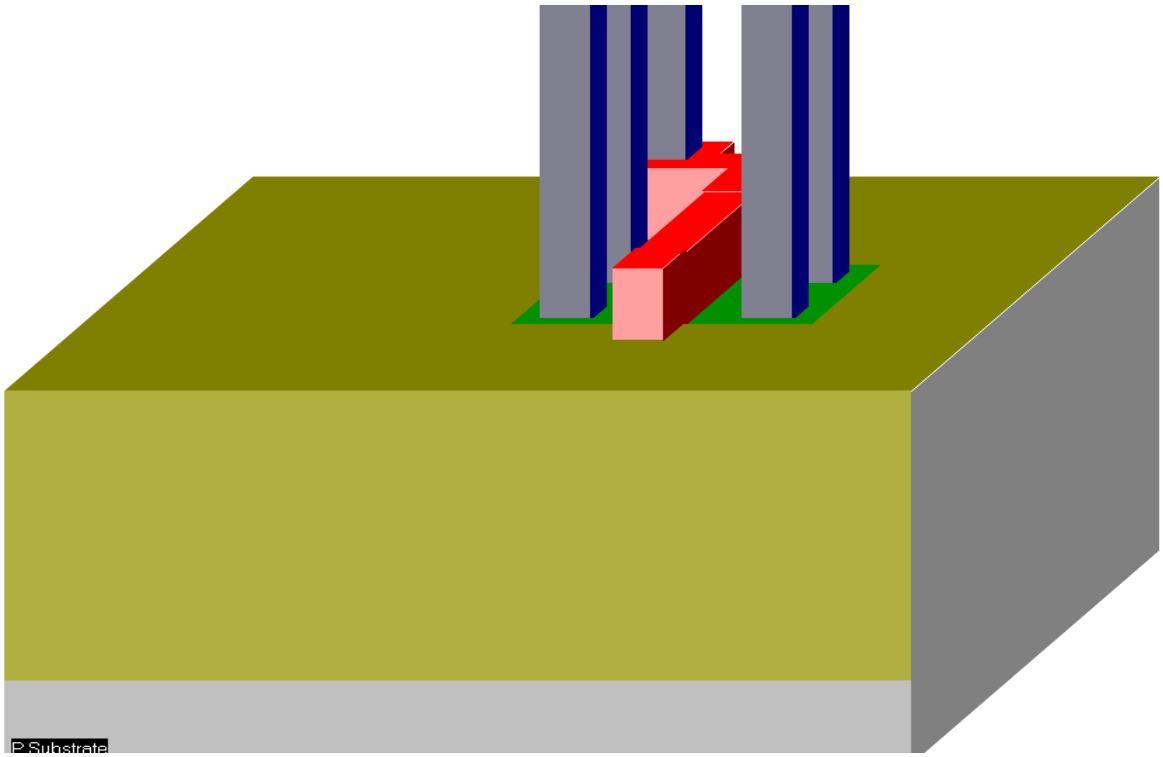


Output:



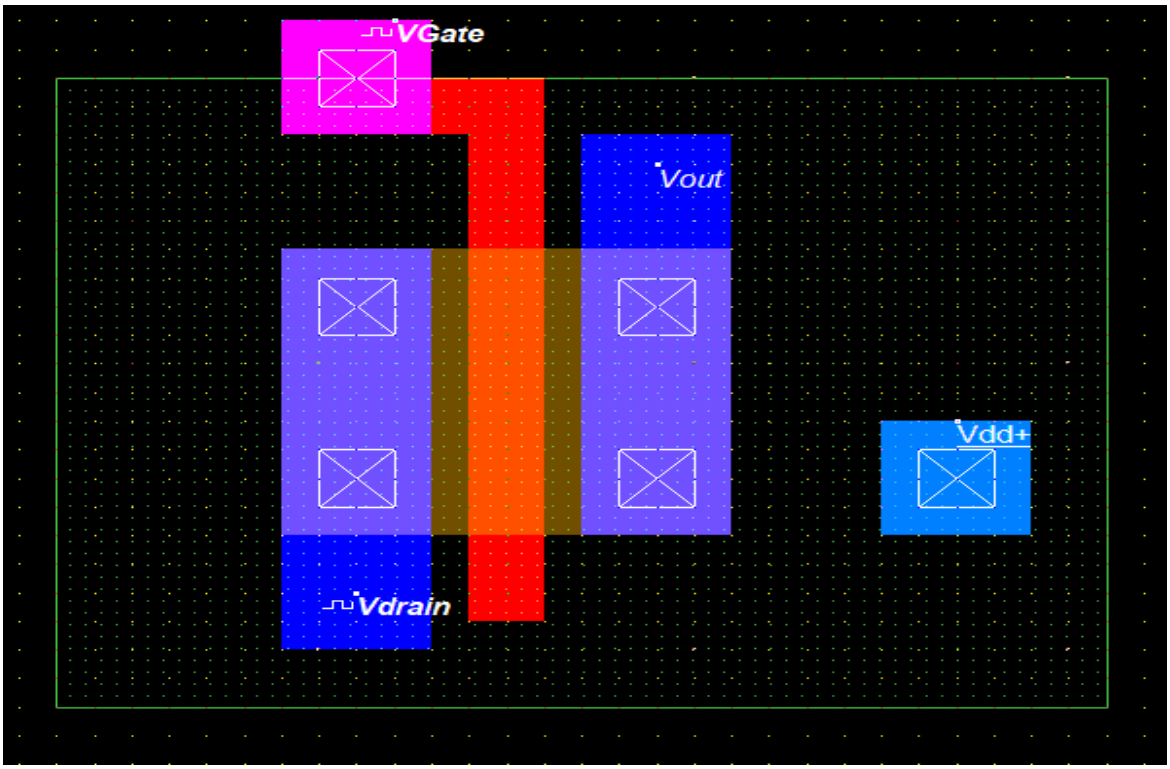


NMOS 3D Process View

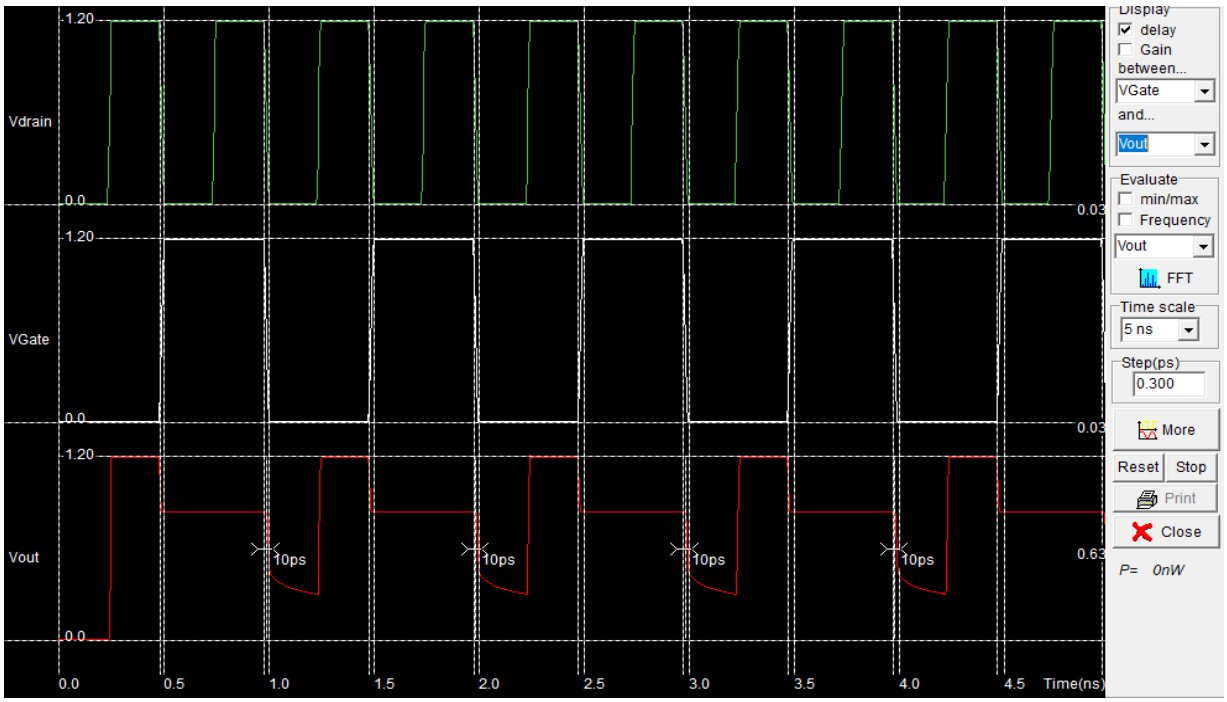


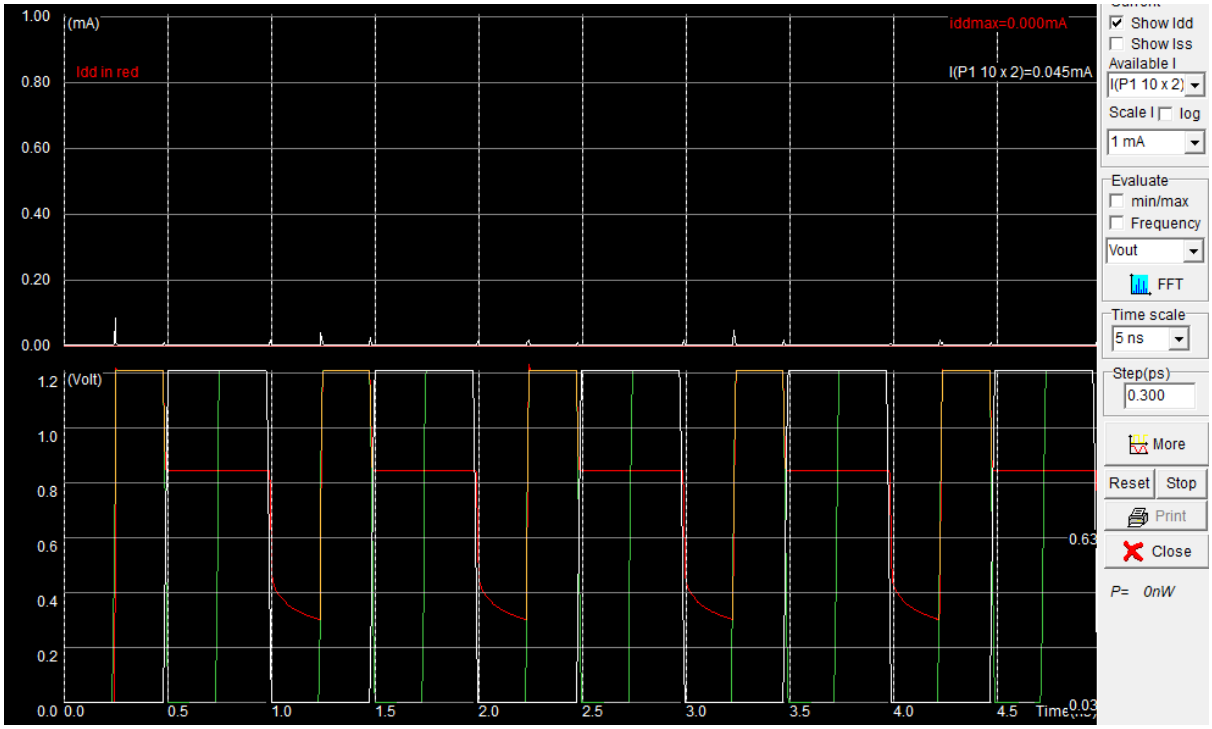
Layout:

P-MOS

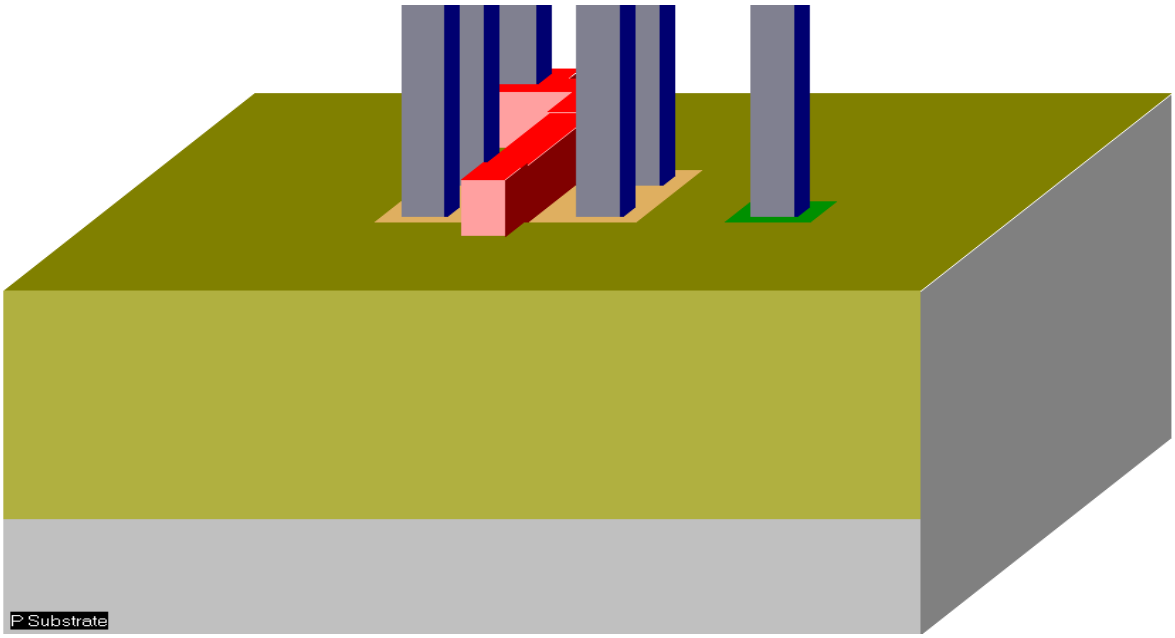


Output:





PMOS 3D Process View



Pre Lab Questions:

1. What is the LAMBDA width of N-mos Diffusion
2. What is the poly-silicon width of N-mos and P-mos

Post Lab Questions:

3. Why do ICs need small power for their operation?
4. Why are ICs more reliable than discrete circuits?

Conclusion: Thus, we have designed and simulated Layout of NMOS and PMOS devices in micro-wind.

CO's Covered:

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